AI use record - Team 4

Draft for the AI sheet (Annex 7) | 29 Oct 2026 | From work/AI-LOG.md | No personal data

AI tools used

- Claude Code (event account, Opus 5.5): code, data checks, web research, PDFs, map app build.
- Teammate's Streamlit app: AI use not recorded here - the teammate adds their own entries.

1. How we used AI at each stage (draft levels - team to confirm)

- 1 Understanding the case and data - AI and team together. Checked our prep for personal data; turned the checklist into a team overview built from Annexes 1, 3, 4, 5; assessed the GEV idea against what the data contains.
- 2 Loading and preparing data - AI did most of it. AI read the real layer and column names from the files before writing code (all EPSG:4326; OBAT fields id, use, height, epoch...).
- 3 Analysis and code - AI did most of it, team directed. Script comparing the observed 2021 flood with EFAS 10/100/500-year zones (buildings, people), reusing our tested helper functions.
- 4 Maps and charts - AI and team together. Map layers and colours; teammate's depth classes reused.
- 5 Desk research and sources - Team led, AI assisted. AI found leads (Walloon Gov press release, Copernicus, Annexe F GEV fits, river-flow and soil datasets); every figure used was opened at the source; unopened sources marked 'verify'.
- 6 Interpreting results - Team led, AI assisted. AI flagged that the first building count was implausibly low and why (Annex 2 limits).
- 7 Recommendations and brief - not started yet.
- 8 Policy tool - AI and team together. Merged the teammate's precomputed data into our offline map template; web page + Android app built; 4-step story for the demo. v0.2: one app with two roles; residents get the essentials (quoted from the Crisis Centre) at every open, flood-zone layers, and a water/electricity/gas status per municipality; policy makers send code-verified DEMO messages that arrive as phone notifications via a laptop relay. No shelter points or statuses invented.

3. Where the AI got it wrong (5 for the sheet)

- Analysis - Labelled the Belgian flood area as 'Liege / Vesdre'; about 90% of the mapped flood is the Meuse north of Liege. Noticed by splitting flooded area by latitude.
- Analysis - Reported 158 flooded buildings before a plausibility check (official: ~48,000 in Wallonia). Noticed by comparing with the Walloon Government figure and Annex 2 limits.
- Desk research - Search summary gave deaths 31-42 and a 271.5 mm rainfall figure without a primary source. Noticed by opening the Walloon Government press release.
- Understanding - First team overview cited annex numbers without reading the annexes. Noticed once the annexes were unzipped and read.
- Policy tool - Used a map setting the app ignores ('opacity' instead of 'fillOpacity'). Noticed by reading the app code before reuse.

Also logged: a wrong data path in the first script run (crashed, fixed).

4. Three key prompts (team to pick - candidates)

- "Can the teammate's GEV / extreme value idea be used?" - turned an idea into a data check: no time series in the data, but EFAS return-period maps and Wallonia's official GEV fits exist.
- "Check the teammate's version and integrate it into ours for Android" - reused their precomputed data instead of redoing it.
- "Overview of what happened in 2021 in Wallonia, detailed" - background with every key figure traced to a primary source.

5. How we verified the results

- Helper functions: 15 automated tests pass before use.
- Two independent methods for buildings in the 2021 footprint (Liege/Meuse area): teammate's pipeline 156, our script 158.
- Official figures only from opened primary sources (Walloon Gov press release 4 July 2022, Copernicus EMSR518 page).
- Column names and file contents read from the data, never guessed. Gauge locations from the official AQUALIM geoservice.
- App: relay tested (wrong code refused, bad input refused); every page element cross-checked; APK contents inspected (version, permissions, localhost-only http).
- Still to do (Friday before 11:00): second Coder recomputes every number shown in the tool and brief.

6. Limits and uncertainty

- 2021 flood map = satellite snapshot (15/17 July), not the peak; misses water in narrow urban valleys (Vesdre).
- OBAT undercounts Belgian buildings in flood areas: building counts are a lower bound. OBAT 'use' codes not yet decoded.
- EFAS excludes rivers under 500 km2 catchment; model run on 1990-2016 data. No own GEV fit (no time series in the data).
- Annexe F peaks and return periods were read off charts by eye: approximate. Meuse gauges not mapped (no official coordinates).

2. Biggest help / biggest letdown (team to write)

To be written by the team in 3-4 sentences. Suggested angle: help = fast data checks and merging two codebases; letdown = confident first framing (Vesdre) and numbers before plausibility checks.